

Her health, understood — in her own language.

PCOS and breast screening, cycle tracking and an AI companion that speaks Urdu and English — private, on her phone, and ready to hand to her doctor.

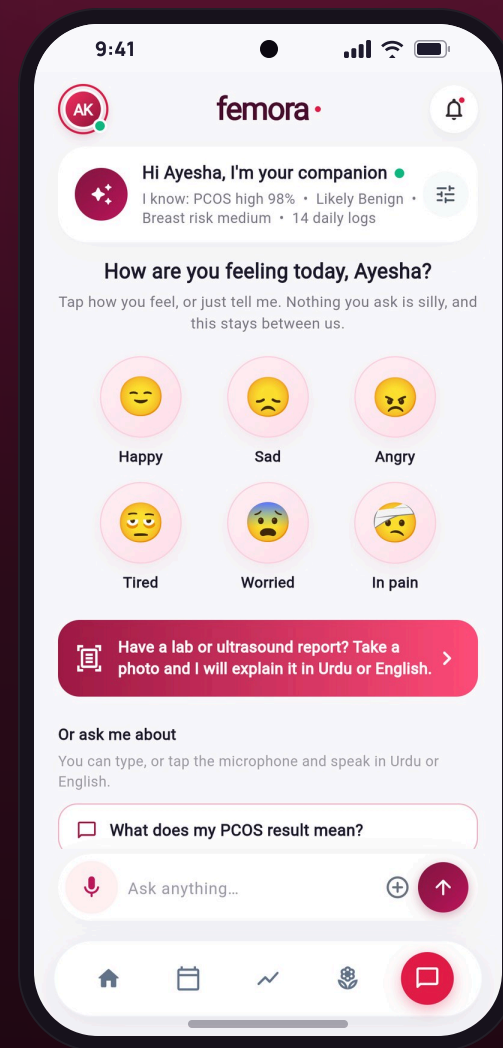
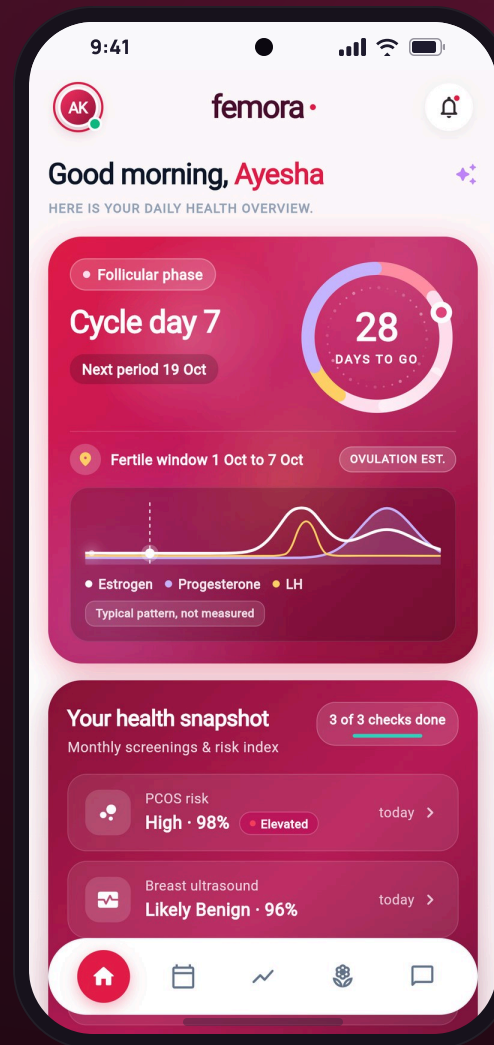
PCOS risk model

Breast ultrasound AI

Cycle prediction

Urdu & English voice

Doctor-ready report



One app for her whole health journey — from first period to the doctor's desk.

Built for women in Pakistan who wait too long for answers. Four jobs, one private app, the same health record on every screen.

SCREEN

Early warning signs, checked

Three trained models look for PCOS and breast-cancer risk from answers and scans.

- PCOS risk (XGBoost)
- Breast ultrasound (ResNet50)
- Breast risk questionnaire

TRACK

Her cycle, learned

Logging, a phase calendar and predictions that learn from her own history.

- Cycle and fertile window
- Symptoms and mood
- Hormonal insights

UNDERSTAND

Answers she can follow

An AI companion that talks and listens in Urdu or English.

- Voice in and out
- Lab report explainer
- Emergency red flags

CONNECT

Care, one tap away

A lab-style report, a QR for the doctor and nearby care.

- PDF health report
- Show my doctor QR
- Find a doctor · nearby care

Symptoms get brushed aside. Femora gives her a clear next step.

What women's health often looks like today — and what it looks like with Femora on her phone.

THE OLD WAY

- ✗ Irregular periods and pain are called "normal" for years
- ✗ Health questions feel too private to ask anyone
- ✗ Lab and ultrasound reports arrive full of unexplained terms
- ✗ Period dates live in memory or a paper diary
- ✗ Every doctor visit starts again from zero
- ✗ Most apps are English-only and made for other countries

WITH FEMORA

- ✓ A PCOS risk check and breast screening in minutes, on the phone
- ✓ A private companion to ask anything, by voice, in Urdu or English
- ✓ Reports photographed and explained in plain words
- ✓ Cycle, symptoms and mood tracked with predictions
- ✓ One tap builds a lab-style report and a QR for the doctor
- ✓ Built for Pakistan: Urdu, local doctors, 1122 emergency

From "is this normal?" to a doctor with her full picture — before lunch.

09:00**Logs her day**

Cramps, low mood, day 7 — logged in a few taps on the cycle calendar.

09:05**PCOS check**

A short questionnaire; the model returns a risk band and what drives it.

09:15**Asks, in Urdu**

She speaks her question; the companion answers aloud, knowing her results.

09:30**Report explained**

A photo of last month's blood test, values flagged against their ranges.

09:40**Finds a doctor**

Gynaecologists near her, with experience, fees and ratings.

11:30**Shows her doctor**

One QR opens her full health report on the doctor's screen.



A PCOS risk check in two minutes — and what she can change.

A short questionnaire about her body, cycle and lifestyle goes to a trained model that returns a risk band and the factors behind it.

1 Trained and tested

XGBoost on self-reported features: 84.4% accuracy, 0.88 ROC-AUC on held-out data.

2 Shows the "why"

BMI, hair growth, weight gain — the drivers appear as tags under the gauge.

3 What-if simulator

Live sliders show how her risk could respond to realistic lifestyle changes, within safe limits.

NOT A DIAGNOSIS A risk-awareness estimate; every result points her to a gynaecologist or endocrinologist.

9:41

< PCOS Risk Check

Answer a few questions about your body, cycle and lifestyle. It takes about 2 minutes.

About You

Age 27 years Height 162 cm

Weight 68 kg

Optional — improves accuracy

Waist Hip

Your Cycle

How regular are your periods?

Regular Irregular

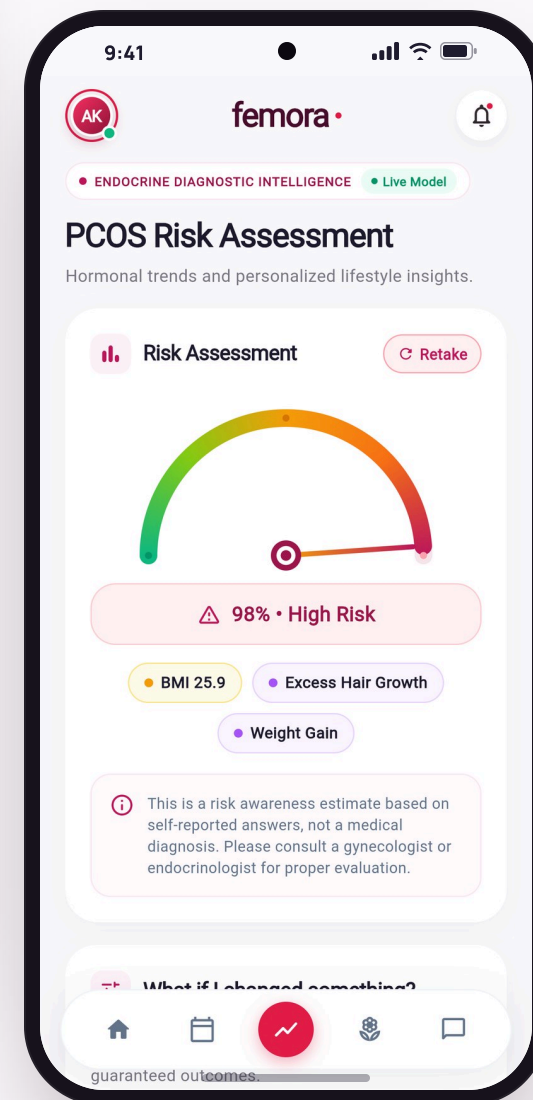
Irregular = cycles shorter than 21 or longer than 35 days, or varying a lot month to month.

How many days does your period usually last?

— 4 days +

Symptoms

Select any you have noticed in the past 6 months



An ultrasound, checked by AI — with the area it looked at outlined.

She uploads a breast ultrasound image. A ResNet50 model classifies it, and a second model outlines the area the result is about.

1 Tuned to catch cancers

At its screening threshold it caught 89.2% of cancers on 710 held-out scans (AUC 0.87).

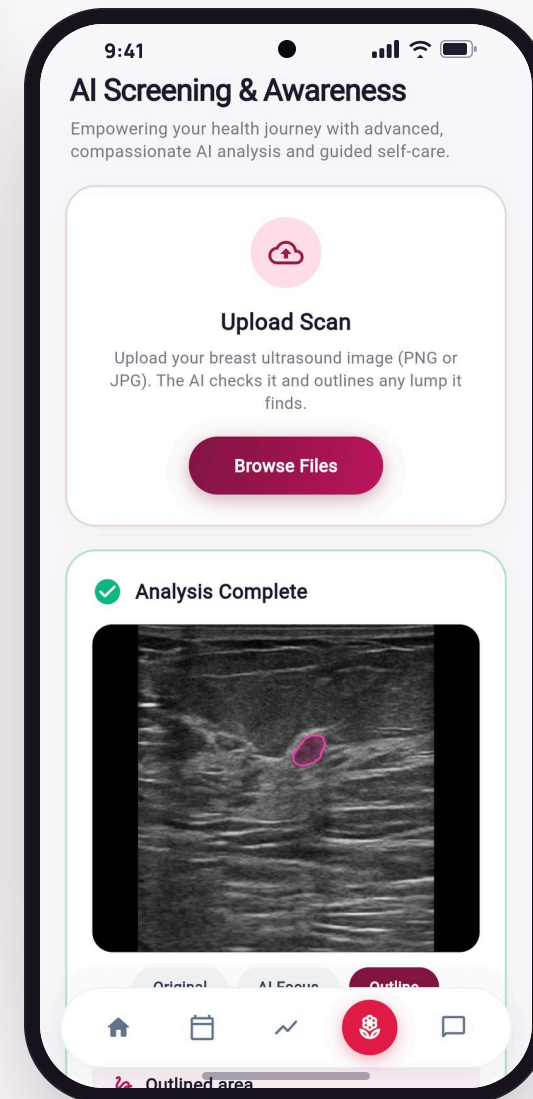
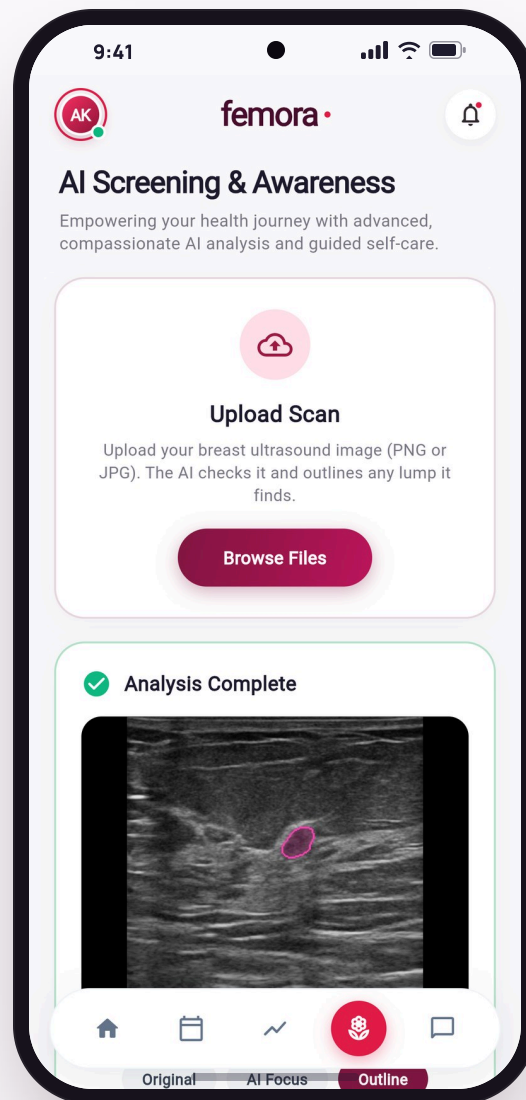
2 Three views

Switch between the original scan, the AI focus heatmap and the lesion outline (U-Net, Dice 0.86).

3 Refuses the wrong picture

A gate checks it really is a breast ultrasound before any result is shown.

SCREENING AID It supports, never replaces, a radiologist's reading.



Know her risk, and know her normal.

A three-minute questionnaire estimates breast-cancer risk; a guided self-exam and a monthly reminder help her notice changes early.

1 Trained on real screening data

XGBoost on the BCSC risk data — 2.39 million mammograms — and well calibrated.

2 Step-by-step self-exam

Seven illustrated steps: when, how long, and what to look for.

3 A reminder that fits her cycle

Once a month, a few days after her period ends.

Her cycle, predicted from her own history.

Log a period in one tap. Femora shows the phase, the next period, ovulation and the fertile window — and says when a cycle looks late or irregular.

1 Learns from her

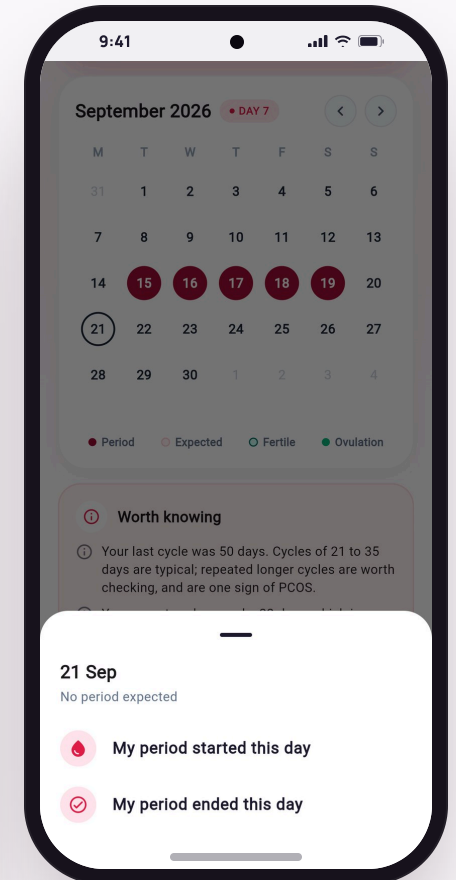
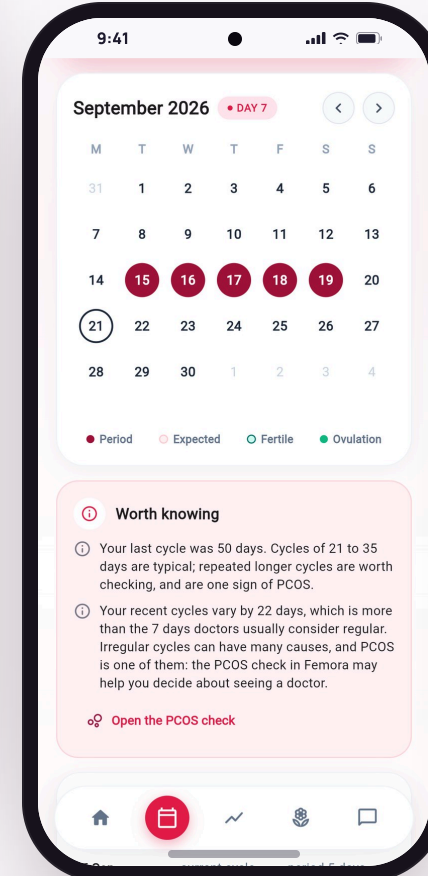
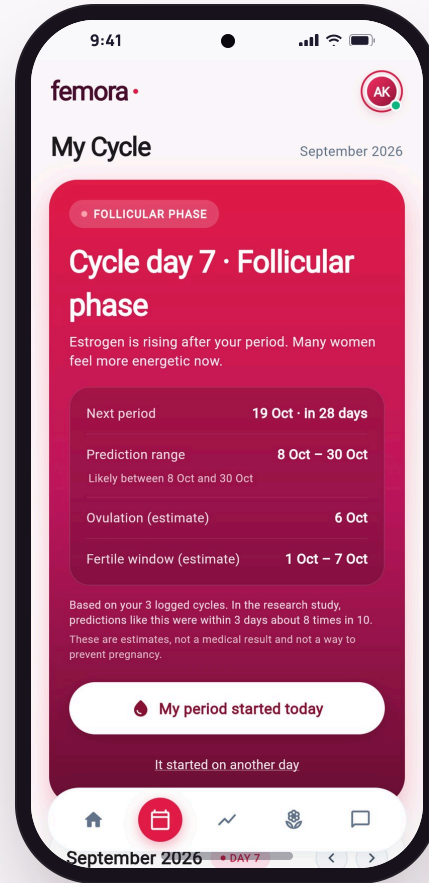
Her own average blended with a study average; after three cycles the next period was within 3 days about 82% of the time.

2 A calendar she can read

Period, expected, fertile and ovulation days in one month view, with a note when a cycle runs long or irregular. Tap any day to log when a period started or ended.

3 Honest estimates

Ovulation and fertile days are clearly marked as estimates, not contraception.



Everything that matters, the moment she opens the app.

A living cycle card and a health snapshot of her latest screenings — all read from one private store on her phone.

1 Cycle at a glance

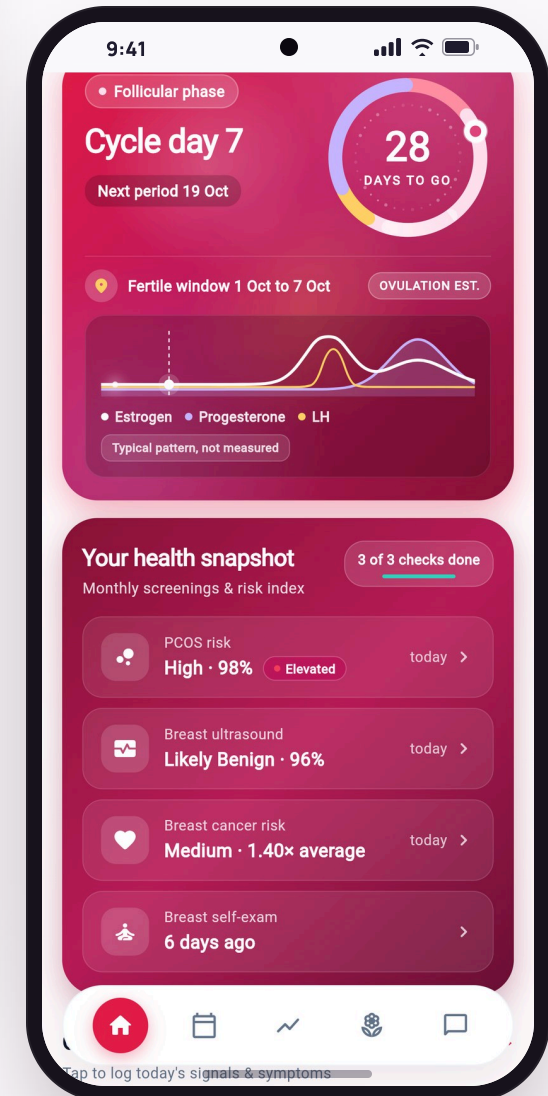
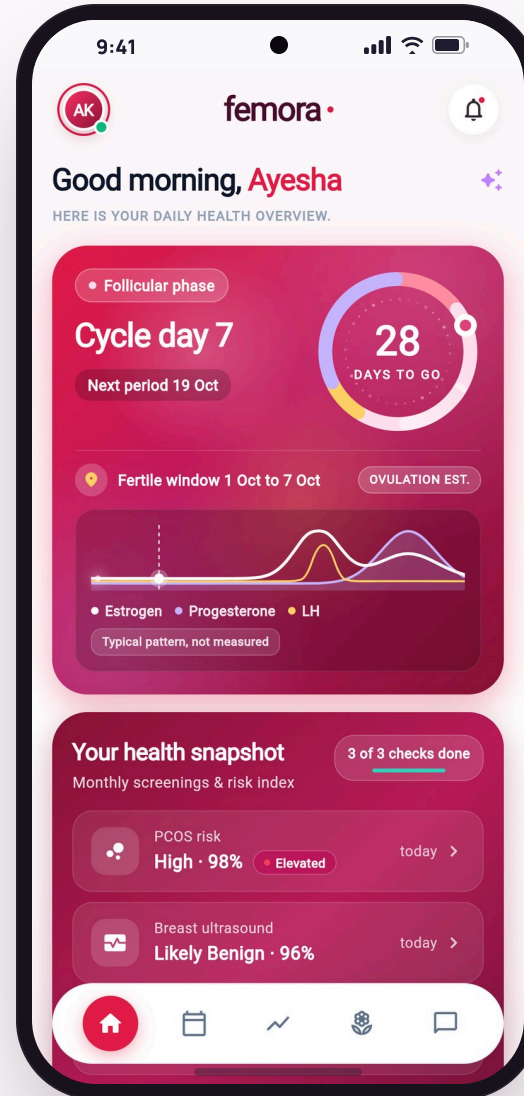
Day, phase, days to go and the fertile window, with a hormone curve.

2 Health snapshot

PCOS, breast ultrasound, breast risk and self-exam — each with its date.

3 One tap deeper

Every row opens its full result or the next step.



Patterns she can finally see.

Symptoms, mood, sleep, stress and energy — logged daily, then laid out against her cycle.

1 Log in seconds

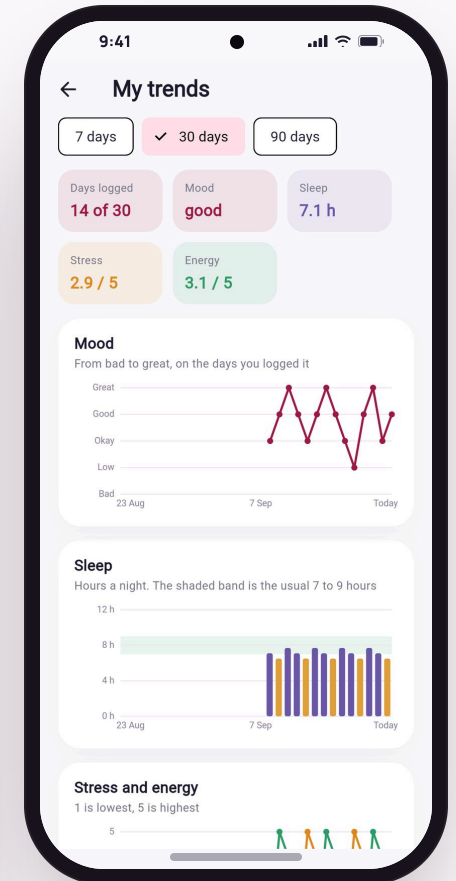
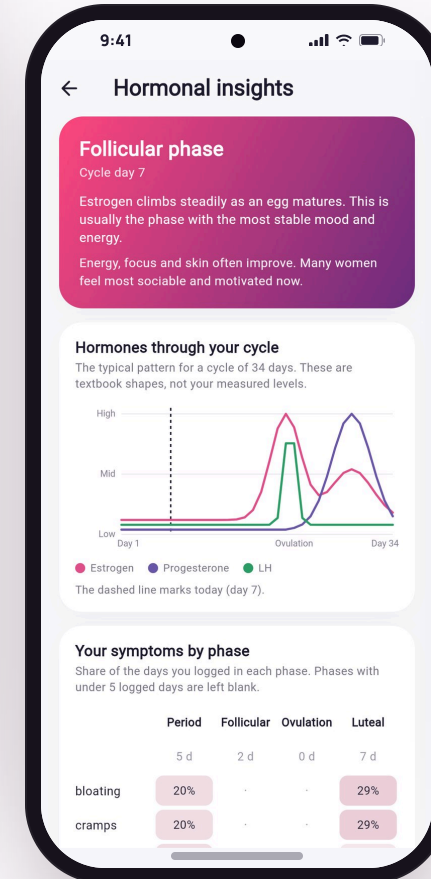
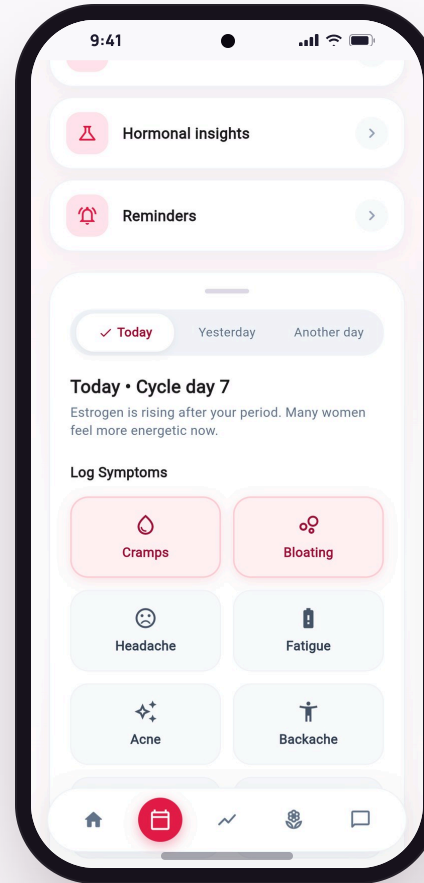
Tap cramps, bloating, headache, fatigue and more for today, yesterday or another day.

2 Hormones through her cycle

Oestrogen, progesterone and LH, with today marked on the curve.

3 Symptoms by phase, trends over time

Which symptoms come with which phase, and mood, sleep, stress and energy over 7, 30 or 90 days.



A companion she can talk to — out loud, in Urdu.

She taps how she feels or asks anything. The companion answers in Urdu or English, by voice or text, and knows her own results.

1 Voice in, voice out

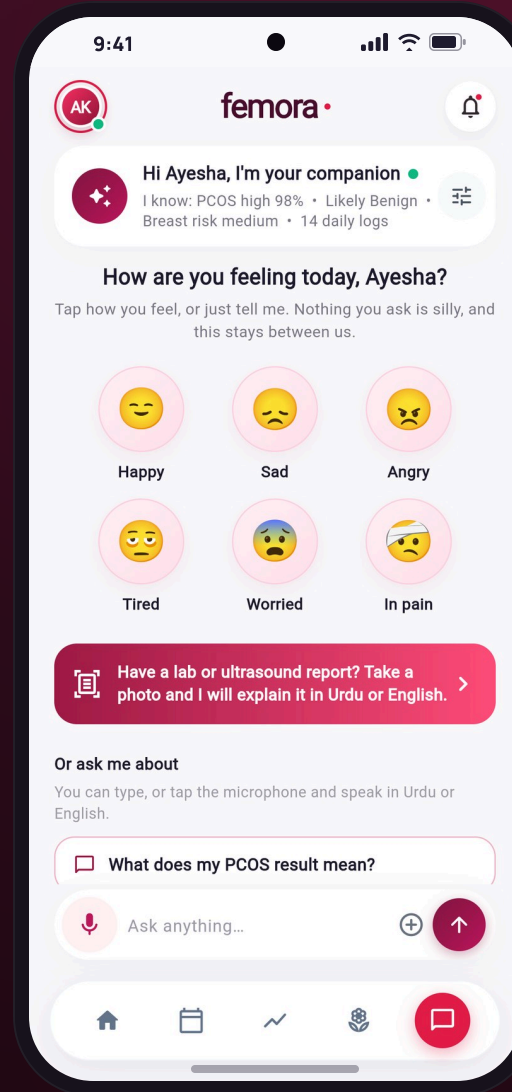
Speech-to-text and a natural voice; the phone's own voice takes over if needed.

2 Knows her context

Her results guide the answers — her name is never sent to the model.

3 Emergency red flags in code

Heavy bleeding and other danger signs are caught in English, Urdu and Roman Urdu, not left to the AI.



Reports she can read — and one she can hand over.

Photograph a lab test, hormone panel or ultrasound report for a plain-language explanation. One tap builds her own lab-style health report.

1 Values against their ranges

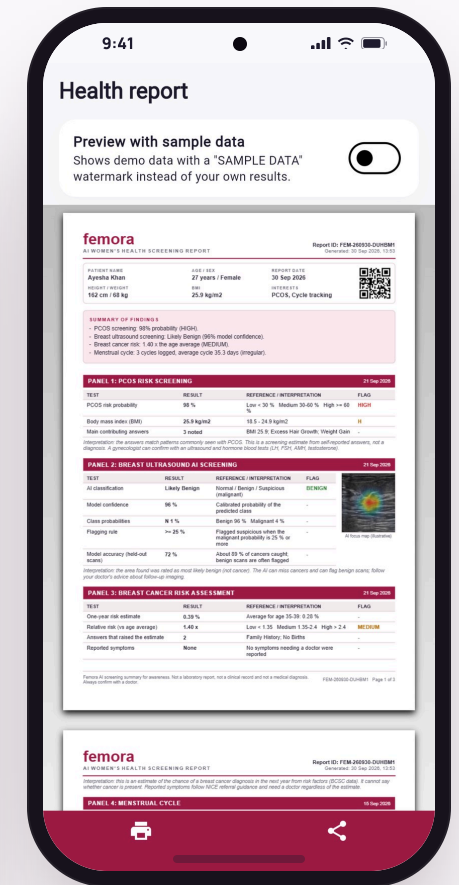
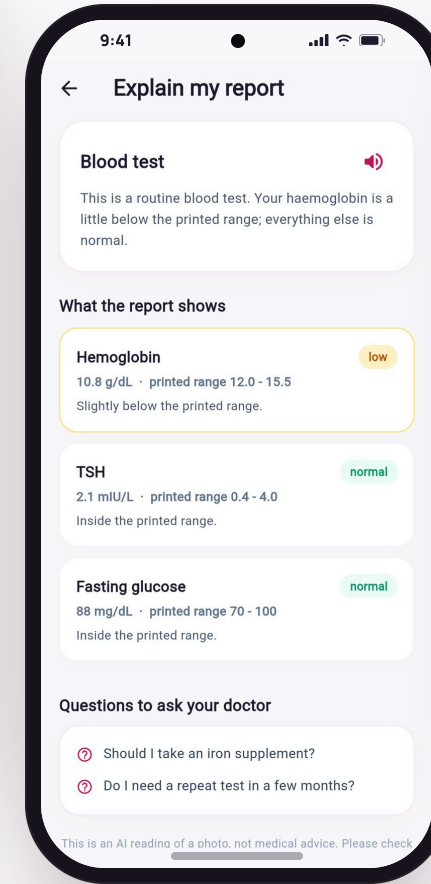
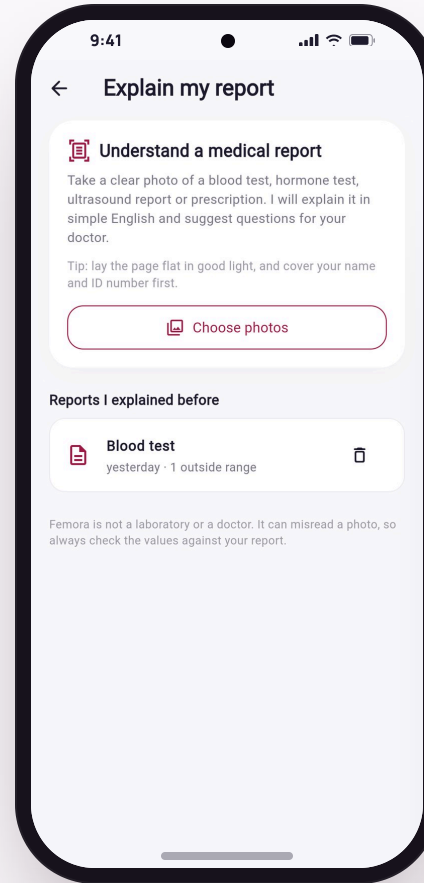
Colour-coded results and questions to ask her doctor, in Urdu or English.

2 Private by design

Photos are cleaned of location data, never stored on the server or kept on the phone.

3 A PDF, one panel per test

PCOS, breast screening and cycle results, ready to print or share.



The right doctor, with her whole picture.

Find a women's-health doctor nearby, then share her results by letting the doctor scan one QR code.

1 Find a doctor

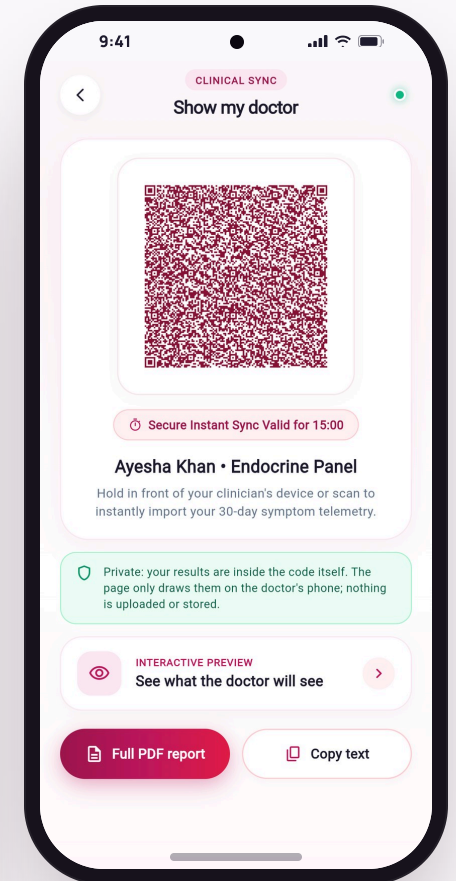
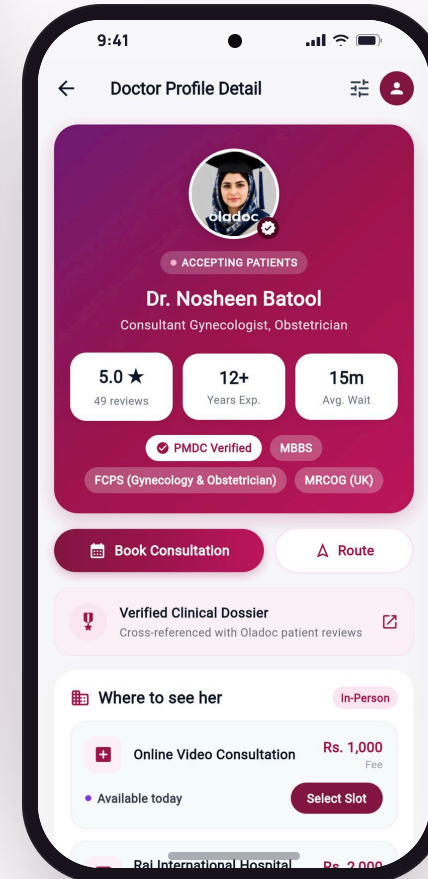
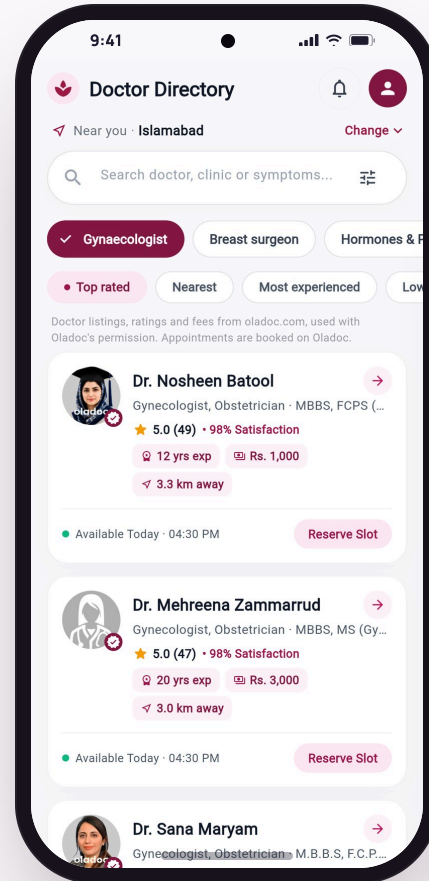
Gynaecologists and breast specialists with qualifications, experience, fees and where to see them (Oladoc listings, used with permission).

2 Show my doctor

A QR opens her full report on the doctor's screen — no app or account needed.

3 Private link

The results travel inside the code itself; nothing is uploaded or stored.



Help nearby, and small daily checks.

Hospitals, gynaecologists, clinics and labs on a map; a morning heart check with the phone camera; reminders that follow her cycle.

1 Nearby care

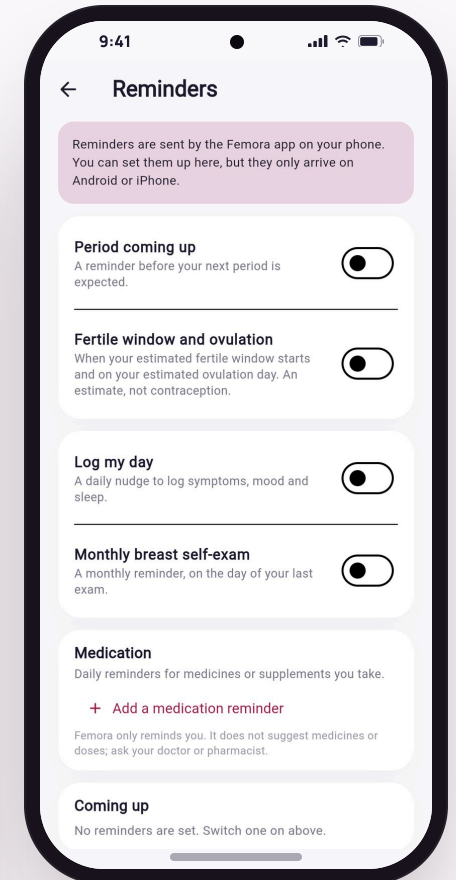
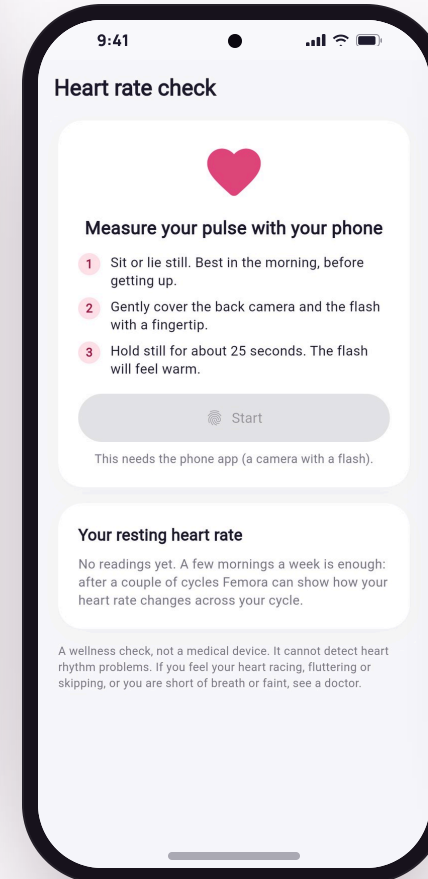
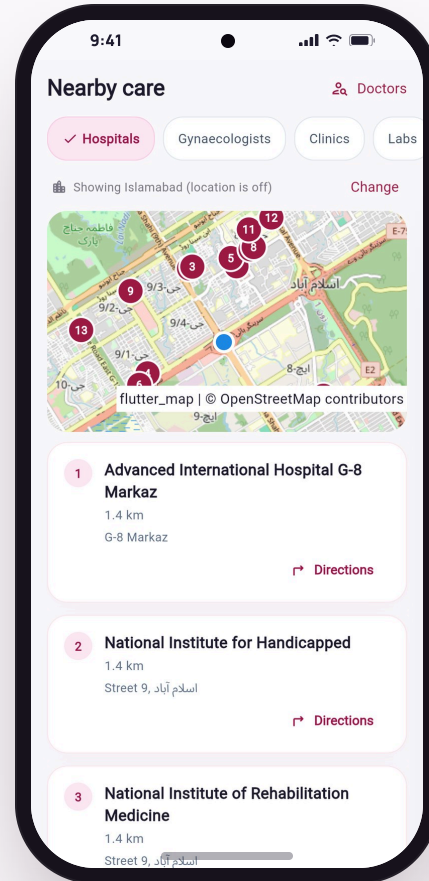
Closest hospitals and clinics with one-tap directions, and 1122 for emergencies.

2 Morning heart check

A fingertip over the camera and flash reads her resting pulse — a wellness check, not a medical device.

3 Cycle-aware reminders

Period, fertile window, daily log, monthly self-exam and medication.



Urdu isn't a translation layer. The whole app turns around.

Switch to Urdu and every screen flips right-to-left, with a bundled Nastaliq font so the text reads the way she writes it.

1 Right-to-left, everywhere

Home, calendar, questionnaires, results and the companion all mirror their layout.

2 Nastaliq built in

Noto Nastaliq Urdu ships with the app, so no phone is left with broken script.

3 Voice in Urdu

She can speak and listen in Urdu, not only read it.



Set up in a minute, on her terms.

A short onboarding asks only what helps — every field is optional, and it all stays on her phone.

1 She chooses what matters

PCOS, cycle tracking, pregnancy or breast health shape what Femora shows first.

2 Her data, her device

One on-device health store feeds every screen; nothing is uploaded to build her profile.

3 Delete everything

"Delete all my data" erases it in one step, and personalisation can be switched off.

femora

Your personal women's health companion. Tell us a little about you so Femora can personalise what it shows and says.

About you

All optional. This stays on your phone.

Name (shown on your report)
Ayesha Khan

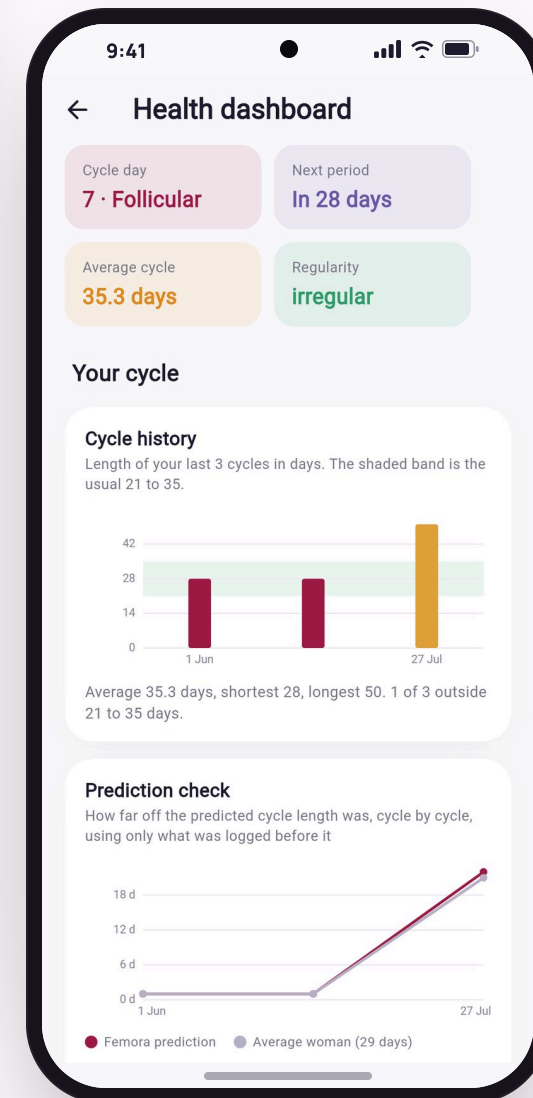
Age 27 years Height 162 cm

Weight 68 kg

What matters to you?

Choose any that apply.

- ☒ PCOS or irregular periods
- ☒ Tracking my cycle
- ☐ Pregnancy
- ☐ Breast health



Designed to help her doctor — never to replace one.

SCREENING

Risk, not diagnosis

Every result is a risk estimate with a clear "see a doctor" next step.

RED FLAGS

Emergencies caught in code

Danger signs trigger an urgent message and 1122 — the AI is not trusted with it.

MEDICINES

No doses, no prescriptions

The companion and report reader never diagnose or give doses.

HONESTY

Estimates labelled as estimates

Ovulation, heart rate and hormone curves say exactly what they are.

On-device store

Her health record lives on her phone.

Name never sent

The AI sees results, not who she is.

Photos not kept

Report photos are cleaned and discarded.

One-tap delete

All her data, gone in one step.

A native app on real models — measured, not guessed.

Flutter on the phone, a FastAPI server for the models, every model trained on public medical datasets and tested on data it never saw.

4

AI models live: PCOS, breast ultrasound, breast risk and lesion outline

400

Flutter tests passing across 41 test files

26.3 MB

Android release APK (arm64), split per ABI

2

Languages end to end: Urdu (right-to-left) and English

Model	Data	Held-out result
PCOS risk · XGBoost	Self-reported symptoms and body measures	84.4% accuracy · ROC-AUC 0.88
Breast ultrasound · ResNet50	BUSI, BrEaST and BUS-BRA scans	AUC 0.87 · 89.2% of cancers caught (710 scans)
Lesion outline · U-Net	Radiologists' outlines	Dice 0.86
Breast risk · XGBoost	BCSC — 2.39 million mammograms	AUC 0.64 (typical for risk models: 0.6–0.7), calibrated

EVERYTHING IN ONE LIST

Every feature in Femora.

SCREENING	CYCLE	TRACKING	COMPANION	DOCTORS & CARE	PLATFORM
PCOS risk questionnaire	One-tap period logging	Symptom logging	Chat in Urdu or English	PDF health report	Urdu right-to-left
PCOS risk gauge and drivers	Phase calendar	Mood check-in	Voice in and out	Show my doctor QR	Nastaliq font bundled
What-if simulator	Next period prediction	Hormonal insights	Knows her results	Web report for doctors	On-device health store
Breast ultrasound AI	Ovulation estimate	Symptoms by phase	Emergency red flags	Find a doctor	Delete all my data
AI focus heatmap	Fertile window	Mood, sleep, stress, energy trends	Lab report explainer	Nearby hospitals & labs	Guest, email, Google and phone sign-in
Lesion outline	Late / irregular notes	Health dashboard	Questions for the doctor	1122 emergency	Android release build
Ultrasound gate	Cycle history	Morning heart check	Personalise on / off	Directions	
Breast risk questionnaire	Prediction check	Cycle-aware reminders			
Self-exam guide					

WHO BUILT IT

A final-year project from Air University, Islamabad.

Femora was designed and built as a final-year project: a phone app, a model server and four trained models, tested end to end.

- **Research-backed**
Every model trained on public medical datasets and evaluated on unseen data.
- **Built for Pakistan**
Urdu, local doctors, local emergency numbers and local privacy expectations.
- **Ready to pilot**
An Android release build and a web report for doctors.



Arshia Naseer

Team member



Ali Haider Bilal

Team member



Ammad Sajjad

Team member



Mustabshera Fatima

Supervisor



Care that every woman **deserves** — in her pocket.

Screening, tracking and answers in her own language, and a clear path to the right doctor.

- ✓ PCOS and breast screening with trained, tested models
- ✓ Cycle predictions that learn from her own history
- ✓ An AI companion that speaks Urdu and English
- ✓ A doctor-ready report, one tap away

